Topology-Correcting Differentiable Triangle-Soup Reconstruction via Persistent Homology

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Abstract

Resampling priors can steer where a differentiable triangle-soup reconstruction spends its triangle budget, but our controlled study of that channel (supplementary material) found their topological value carried largely by spatial width, and no prior shape repairs one-dimensional features: concentrated priors manufacture phantom handles, spread priors never beat a random control. We move topology from the allocation channel into the objective: L = photometric + $\lambda(t)\, L_{\text{topo}}$, where L_{topo} is an optimally-matched distance between the live surface's persistence diagram and a fixed target diagram. Gradients flow through a pair-frozen backward pass: matched birth/death simplices are re-expressed as closed-form circumradii of live sample positions — exact for Gabriel simplices, which constitute all significant pairs in practice — and a recruitment term restores the gradient optimal matching provably lacks when a target feature is missing. A single ratio knob ρ sets the loss scale by gradient-norm calibration; a curriculum proved unnecessary. On budget-starved synthetic scenes the loss is topology-specific where a norm-matched non-topological control is not: enclosed voids improve 4.0–7.9× and — the channel's distinguishing result — torus loops improve 2.3× with zero phantom handles across seeds, while the control collapses a loop. The channels compose: loss plus the best prior reaches 7.6× below baseline at the best Chamfer. Component counting (H0) is again matched by the generic control — this class needs no topological information. All verdicts replicate, with no per-shape tuning, on four external genus-known meshes — one a raw artist mesh ingested and certified rather than constructed — at 2.1–10.4× and equal-or-better Chamfer, and the loss degrades gracefully (never below baseline) under sensor noise up to one sample spacing. Prescription: correct topology in the loss, allocate with a wide prior, and combine.

1. Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [41], periodically resampling triangles under a fixed budget. Photometric losses are topologically blind: reconstructions that differ in genus, connectivity, or enclosed-void structure can be indistinguishable to Chamfer and Hausdorff while the bottleneck distance between persistence diagrams separates them by an order of magnitude (suppl. Table S1). We first attacked this through the allocation channel — biasing where the resampler spends triangles — in a controlled study reported in the supplementary material (§A). The verdict was mixed: a topological prior helps enclosed voids, but most of the gain is reproduced by any equally-wide non-topological prior, and no prior shape repairs one-dimensional features: concentrating manufactures phantom handles, and spreading never beats a random control on loops.

This paper moves topology into the objective: a differentiable persistence term in the training loss, so gradients act on the birth/death values of the reconstruction's topological features directly, rather than on where triangles respawn. Concretely:

- A matched persistence loss with a pair-frozen backward (§3): the forward pass is ordinary exact persistence (GUDHI alpha complex); the backward pass re-expresses each paired birth/death simplex as the closed-form circumradius of live surface samples; 100% of significant pairs are Gabriel-exact on our shapes, so the frozen gradient is the true gradient almost everywhere.
- A recruitment term for missing features: optimal partial matching yields exactly zero gradient when a target feature has no live counterpart — the most important failure mode. Recruitment attaches the nearest live bar instead, restoring a descent direction; a controlled toy probes its known location-blindness, and an in-loop ablation shows the term is load-bearing — removing it collapses the loop-class

win to baseline (§5).

- One calibrated knob: λ is set by a gradient-norm ratio ρ against the photometric loss; the response is flat over a 10× range, and a curriculum adds nothing once calibrated — an ablation, not an assumption.
- Channel-controlled evaluation: every claim is tested against a norm-matched non-topological control acting through the identical channel (a repulsion on the same surface samples), inheriting the allocation study's control discipline.

The loss is topology-specific for enclosed voids (4.0–7.9× at Chamfer parity) and, unlike every prior we tested, for loops (2.3×, zero phantom handles). The channels compose (best-of-both 7.6×); components (H0) are again matched by the generic control — three channels agree the class carries no exploitable signal.

2. Related work

A decade of gradient-based reconstruction: four positions on connectivity. OpenDR [28], neural mesh rendering [22], SoftRas [27], DIB-R [7] and nvdiffrast [24] established gradient flow from pixels to geometry; reconstruction work since splits on what that geometry should be. (i) Implicit fields — learned occupancy and signed distance [30, 35], volume-rendered from NeRF [31] through NeuS [43] to Neuralangelo [26] — deliver watertight, topologically consistent surfaces by construction — the strongest argument against explicit soups — paid for with dense field evaluation and post-hoc extraction. (ii) Fixed-connectivity templates deformed per scene [16, 42] keep manifoldness but freeze topology at the template's genus. (iii) Differentiable iso-surfacing re-extracts the mesh from a volumetric scaffold at every step [32, 39, 40] (Shape-as-Points bridges point soups to watertight implicits [36]) — topological freedom at grid cost. (iv) Connectivity-free primitive soups — point splatting [44], patch atlases [14], Gaussians [23] and their surface-accurate variants [15, 20], triangle splatting [17], and DiffSoup [41] — are the fastest and the easiest to budget, but even their surface-minded members regularize geometry (alignment, flatness), never a measured topology. (i) and (iii) solve topology representationally; (iv) had not addressed it — the gap this paper fills at the soup's budget and speed. We build on DiffSoup, whose in-loop prune/respawn resampler is exactly the mechanism our allocation-channel study (suppl. §A) steered with importance priors; here the resampler stays untouched and the objective changes. Vertex-domain optimizers such as large-steps preconditioning [33] are complementary — they change how vertices move, we change why.

The allocation channel: why priors alone fail. Three findings of our allocation-channel study set this paper's problem. (1) Geometric metrics are topology-blind: on bisection-matched probes with Chamfer equal by construction, the discriminating-dimension bottleneck separates correct from broken topology by 35–40× (suppl. Table S1). (2) For voids, a spread topological prior cuts the bottleneck 33–53% below baseline — but a width-matched random control recovers most of that gain, leaving a small, shape-dependent residual: the prior's value is carried largely by its width. (3) No prior shape repairs loops: the best torus arm is the non-topological wide control (.0267 vs the topological spread prior's .0316), and the concentrated topological field manufactures phantom handles (4.4 significant loops against the true 2). The composition condition C5 reuses that study's strongest field (B4; suppl. §A) — though any width-matched allocation prior could fill the slot.

Persistence in learning objectives. Persistent homology [12] with its stability guarantees [9] has been used as a trainable signal via differentiable sorting of filtration values [5], for topology-preserving image segmentation [8, 19], point-cloud and shape optimization [4, 37], and diagram-metric optimization with convergence analysis [6]; barcode-valued maps have since been given a general differential framework [25], and the per-step cost of persistence-guided descent attacked directly [34]. Our backward pass is in this family — unlike work that back-propagates through filtration-value sorting or barcode coordinates in general position, we re-express each paired simplex's birth/death as a closed-form circumradius of live surface samples — with three pipeline-specific ingredients. (i) Alpha-complex pairing over $\alpha\times$area surface samples of a live triangle soup, rather than image grids or raw clouds, couples the diagram to the geometry the renderer is actually training. (ii) A Gabriel-exactness gate: the circumradius re-expression equals the filtration value only for Gabriel simplices, so we verify it per pairing (relative 10^{-6}) — orthogonal to the standard pair-frozen device: it certifies the frozen gradient rather than assuming it. (iii) Recruitment for missing features: optimal partial matching has zero gradient exactly when a target feature is entirely absent — the worst failure mode — so we trade metric fidelity for a restored descent direction (§3), quantifying both its pathology and its necessity (§5).

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [2, 21, 38], and budgeted remeshing has a long lineage [1, 3, 10, 13, 18]; our contribution is orthogonal — the budget mechanics stay DiffSoup's, only the training signal becomes topology-aware.

3. Method: a matched persistence loss for triangle soups

3.1. Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary $+\, 0.8\, L_1 + 0.2\,\text{SSIM}$) [41], which we adopt unchanged: every ablation below isolates the topology channel, never photometric tuning. We add

$$L \;=\; L_{\text{photo}} \;+\; \lambda(t)\, L_{\text{topo}}(V), \tag{1}$$

where V are the soup's vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed target bundle $\mathcal{T}$: the significant finite bars of the ground-truth shape's diagram, computed once at the sampling density M the loss will see (alpha values shift with point spacing: density matching is a correctness requirement) and normalized by the target's bounding-box diagonal, locked for all live evaluations.

3.2. Live diagram and pair-frozen backward

At each refresh we draw M surface samples ($M{=}2048$ unless the measurement-floor rule of §4 dictates a denser bundle) $X_j \;=\; \sum_i w_{ij}\, V[F(f_j), i]$ with faces f_j drawn $\sigma(\alpha)\times$area-weighted and barycentric weights w_{ij} frozen; X is recomputed from live V every step, so gradients reach vertices while the combinatorics stay fixed. The alpha complex of X [11], computed exactly with GUDHI [29], yields persistence pairs (birth simplex, death simplex) per dimension. For a Gabriel simplex the alpha filtration value is its squared circumradius, so each paired bar's coordinates are closed-form functions of a few sample positions, all exactly differentiable:

$$\begin{gathered} R_{\text{edge}} = \tfrac{1}{2}\|p_1{-}p_0\|, \qquad R_{\text{tri}} = \frac{\|u\|\,\|v\|\,\|u{-}v\|}{2\,\|u\times v\|}, \\ R_{\text{tet}} = \|y\| \quad \text{with} \quad A\,y = \tfrac{1}{2}\,\text{diag}\big(AA^{\top}\big), \end{gathered} \tag{2}$$

where $u = p_1{-}p_0$, $v = p_2{-}p_0$, and A stacks the rows $p_i{-}p_0$ (the circumcenter is $p_0{+}y$). Edges carry H0 deaths and H1 births, triangles H1 deaths and H2 births, tetrahedra H2 deaths. We verify, per pairing, that the re-expressed value matches the filtration value (relative 10^{-6}); a pair failing this Gabriel test is detached — it keeps its (correct) value in the loss but propagates no gradient through that simplex, so an exactness failure can never inject a wrong gradient; the worst case is a missing tug for one refresh interval. In practice the gate never fires: 100% of significant pairs are Gabriel-exact on all six shapes' clean clouds, with zero failures in the stress test's 2,400 noisy refreshes (§5). The per-refresh failure count is logged, and stronger fallbacks (dropping failing pairs, subgradient surrogates, locally raising M) remain compatible with the gate. The pairing is treated as locally constant — the standard pair-frozen device [5, 6] — refreshed every $K{=}10$ steps and, necessarily, after every resampling event, which replaces the vertex and face tensors wholesale.

3.3. Matching, diagonal pressure, and recruitment

Let D be the live significant bars and $\mathcal{T}$ the target bars of one dimension. An optimal partial matching (Hungarian on the standard augmented cost; equivalently W_2) splits bars into matched pairs, unmatched live, and unmatched target:

$$\begin{aligned} L_{\text{topo}} = &\sum_{(i,j)\text{ matched}} \big[(b_i - b_j^*)^2 + (d_i - d_j^*)^2\big] \\ &+ \sum_{i\text{ unmatched}} \Big(\tfrac{d_i - b_i}{2}\Big)^2 \;+\; L_{\text{recruit}}. \end{aligned} \tag{3}$$

The first term pulls matched features to their target coordinates; the second pushes spurious features to the diagonal. The third exists because optimal matching has a blind spot: when a target feature is entirely missing, the cheapest transport diagonalizes both sides and the gradient is exactly zero. Recruitment restores a descent direction. Let $\mathcal{T}_{\text{miss}}$ be the unmatched target bars, processed in decreasing persistence order, and let the candidate pool $\mathcal{A}$ hold every live bar not claimed by the matching — unmatched significant bars and, crucially, sub-threshold bars, which matching never sees. Each missing target greedily claims its nearest pool bar, without replacement, under the same squared birth–death metric as the matched term:

$$\begin{aligned} L_{\text{recruit}} &= \sum_{j\in\mathcal{T}_{\text{miss}}} \big[(b_{k(j)} - b_j^*)^2 + (d_{k(j)} - d_j^*)^2\big], \\ k(j) &= \arg\min_{k\,\in\,\mathcal{A}_j} \big[(b_k - b_j^*)^2 + (d_k - d_j^*)^2\big], \end{aligned} \tag{4}$$

where $\mathcal{A}_j$ is the pool remaining after earlier recruitments; recruited bars are excluded from the diagonal term. Both coordinates of a recruited bar backpropagate through the same circumradius machinery. This deliberately departs from a metric-faithful W_2; §5 probes its known pathology (location-blindness) directly, and the C6 ablation shows the term is load-bearing in training.

3.4. Calibration instead of curriculum

$\lambda(t)$ ramps linearly from 20% to 50% of training, with peak set once by a gradient-norm ratio at the first active step: $\lambda_{\text{peak}} = \rho \,\|\nabla_V L_{\text{photo}}\|/\|\nabla_V L_{\text{topo}}\|$. The single knob ρ is dimensionless; ρ=0.1 throughout. The tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, and an uncalibrated-schedule ablation (constant λ from step one) matches the ramped version — calibration, not scheduling, is what makes the knob safe. The window carries no measurable sensitivity either: alternative ramps (10–40%, 30–60%; sphere, three seeds) land at .0154±.0017 / .0179±.0019 vs the default's .0156±.0013 — within seed noise; 20:50 is a conservative choice, not a tuned hyperparameter.

3.5. Conditions

C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2: + a short-range repulsion on the same frozen samples, calibrated with the same ρ — a norm-matched non-topological control through the identical channel (C2g: a gentler radius). C3: C1 without the ramp. C5: C1 + the best prior of our allocation study (the spread topological field B4; suppl. §A). C6: C1 with the recruitment term removed (pure matching + diagonal, all else identical) — does recruitment matter in-loop, or only for cold-start repair? The hook adds one loss line to the training loop; flag off, no new code executes, and a ρ=0 run is divergence-equivalent to baseline under CUDA nondeterminism.

4. Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (every-8th test holdout ⇒ 21–63 training views): sphere and cube (void class H2, β=$(1,0,1)$, budget N=1200), torus (loop class H1, $(1,2,1)$, N=700), two spheres (component class H0, $(2,0,2)$, N=700), and genus-2 double torus as a stretch case ($(1,4,1)$, N=2000). Budgets are the allocation study's headroom points (suppl. §A): tight enough that baseline topology is imperfect, so there is something to win. Four external, genus-known meshes probe generality beyond the analytic family (§5.2). Three are watertight benchmarks: spot (genus 0) and bob (genus 1) from the Crane model repository (CC0), and the fandisk CAD benchmark (genus 0) — rendered with the same pipeline (48 views, 200 px) at unchanged class budgets; a seed-0 C0 pilot confirms baseline imperfection (tail bottleneck .043–.054, inside the analytic band). The fourth, the tom-yum pot (Fig. 2), probes the step every real deployment needs: a raw artist-authored mesh rather than a clean benchmark. As exported from Blender it is textbook triangle soup — 6,318 vertices in 9 overlapping open shells, mixed quad/12-gon faces, 50 boundary edges after exact welding, 232 non-manifold junction edges — with no trustworthy ground-truth topology (the same defect that excluded ShapeNet). An ingest-and-certify pass makes it usable: exact-duplicate weld, offset-solidification into a thin closed shell (exact unsigned distance field + marching cubes at iso ε), dropping 804 enclosed pocket shells, then certification by measurement (closed, edge-manifold, consistently oriented; exact simplicial homology; watertightness cross-check; bit-identical rebuilds), yielding one body of genus 3, β=$(1,6,1)$ exact (190,990 vertices), rendered and budgeted as a void-class shape (N=1200, 48 views, 200 px). Where the benchmark trio's topology is known by construction, the pot's is known by certificate — the pipeline real scans will need. The pot also exercises the measurement floor of §6 prospectively: no H1 loop of the certified shell (handle bores, flue) clears the $6\,r_{\text{med}}$ floor at any working M, while the narrow flue mouth caps at small α, making the flue interior an enclosed chamber — an H2 void on the pot axis. The void first clears the floor at M=8192 (margin 1.20–1.23× across five sampling seeds), so the pre-registered floor rule builds its target bundle there — the study's only protocol deviation — and the loss samples the live surface at the same density (the density-matched contract). The bundle carries exactly one significant bar (H2), stable to re-sampling ($\sim$$10^{-5}$).

Protocol. 2,500 steps; resampling every 100 steps (untouched); ρ=0.1, ramp 0.2:0.5, refresh K=10, M=2048; decisive shapes (sphere, torus) run 5 seeds with the full condition set (C0/C1/C2/C3/C5, plus the recruitment ablation C6), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2), double torus 2 seeds. A sensor-noise stress test (C7/C7h = C1 + noise; decisive shapes, 3 seeds) perturbs the cloud the loss plan sees with i.i.d. Gaussian noise at every refresh, σ = 0.5%/1% of the target diagonal (≈0.5/1.0× the median sample spacing at M=2048): pairing, Gabriel gates, and recruitment decide on the noisy observation; the pulls apply to the true live positions. The torus restricts the loss to H1: its own void bar is sub-threshold at M=2048, and a loss must not act on features it cannot reliably measure at its sampling density. For the same reason the double torus targets its two measurable loops (of four) — matching what the reconstruction itself can express at any feasible budget. The generality wave runs C0/C1/C2, three seeds; bob (fat tube, unlike the thin torus) has both loops and void significant at M=2048 and runs the full bundle; the tom-yum pot runs at the floor-rule density above.

Metrics and verdict. Primary: bottleneck distance to the target diagram in the shape's discriminating dimension, tail-averaged over the last trajectory dumps, seed-averaged (20k evaluation samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. Verdict rule: C1 must beat C0 and every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not a topological win. Chamfer parity, formally: the passing arm's seed-mean Chamfer must satisfy $\mathrm{Ch}(\mathrm{C1}) \leq 1.15\,\mathrm{Ch}(\mathrm{C0})$; the 15% headroom absorbs seed variance and CUDA nondeterminism and in practice never did any work — every passing arm's Chamfer was at or below baseline. (CUDA nondeterminism itself is quantified in §7.)

5. Results

A gate first: the loss alone (no photometric term, Adam on point positions) repairs defective clouds on all four controlled probes — a punctured sphere's void closes (8.8×), a spurious handle dies at the diagonal, misspaced components reach target separation (1229×), and — the recruitment probe — a bridged merge with no significant live H0 bar is severed (450×, $\beta_0\,1{\to}2$), the case where optimal matching provably has zero gradient. The predicted location-blind pathology did not materialize (mechanism and figure: suppl. §C, Fig. S5).

5.1. The C-matrix

Table 1 reports tail bottleneck (mean±sd over seeds) in each shape's discriminating dimension; Chamfer in the text where it matters; Fig. 1 shows the full training trajectories for the two decisive shapes.

Voids (H2): topology-specific, large. Sphere 4.0× (16.2σ Welch vs C0), cube 7.9× (18.5σ), both at equal-or-better Chamfer (sphere 0.46 vs 0.50; cube 0.92 vs 1.05). The norm-matched control is not merely weaker — it is destructive: it pins the void's death at its repulsion equilibrium (2.2× worse than baseline, identical across seeds) and on the cube erases the void outright. Nor is this control-strength dependent: a gentle variant at half the push radius still lands 1.6× worse (.0972 vs .0623) — the ordering is C1 < C0 < C2g < C2, so no tested strength reproduces any of the gain (per-seed spread: suppl. Fig. S3; pin-and-drag mechanism: §6).

Loops (H1): the channel's distinguishing result. In the allocation study (suppl. §A) no prior shape won this class: concentration manufactured phantom handles; spreading never beat a random control. The loss cuts torus H1 error 2.3× with zero phantom handles in all five seeds (significant-count trajectories flat at the true β_1=2, suppl. Fig. S4), while the control collapses one loop by step ∼800 and never recovers it. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not: a loop has minimal spatial support, so allocation can only over- or under-tessellate its neighbourhood (manufacturing phantoms or thinning the tube), while value pressure moves, merges, or tightens the loop directly (§6). Voids, whose support is the whole shell, respond to both channels — why the composition (C5) wins everywhere.

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error 9.5×, but the generic control matches it on the diagram metric (.0008 vs .0007); the topological increment appears only in Chamfer (0.54 vs 1.94). Three channels now agree on the feature class: β_0 structure is carried by the count and spatial separation of shells — properties any spreading pressure improves and the photometric term already infers from images. Topological guidance earns its keep where geometry and topology decouple: loops and voids.

Composition (C5). Loss + spread prior is best everywhere tested — sphere .0082 (7.6× vs C0, ≈2× vs C1 alone, ≈4.4× below the prior alone), torus .0133 — with the best sphere Chamfer as well; on the torus its Chamfer (.502) trails only the no-ramp arm (.495). Allocation and value-tuning are complementary mechanisms, not substitutes.

Stretch case (genus 2): saturated, honestly. On the double torus every arm — baseline, loss, control — returns a bottleneck of exactly .0264 (sd 0 over seeds), and Wasserstein is equally pinned (.0527–.0528): the two tube loops are unexpressable at feasible budgets (replicating the allocation study's floor, suppl. §A), and they dominate both diagram metrics regardless of guidance. The loss adds no phantoms (#sig β_1 = 2 every run) at marginally best Chamfer; the case measures the representation's ceiling, not the channel's — both floors are quantified in §6.

Ablations. ρ-response flat over $\{0.03, 0.1, 0.3\}$ (.0181/.0172/.0177 in the matched seed-0 sweep on the sphere; the five-seed ρ=0.1 arm is Table 1's C1); C3 (no curriculum) ≈ C1 on both decisive shapes and off the analytic family (bob: C3 .0171±.0026 vs C1 .0208±.0008) — the curriculum-free finding generalizes. Overhead, honestly: the loss adds a fixed ≈46 s per 2,500-step run (refresh 174 ms at M=2048 every 10 steps) — more than the 33–41 s photometric baseline on these deliberately tiny scenes, but the cost scales with M ($O(M \log M)$ per refresh), not scene

Table 1. Tail bottleneck to target (lower is better); $\times$ = reduction vs C0. Verdict: C1 < C0 and C1 < every control at Chamfer parity.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior
sphere (H2)	.0623±.0063	.0156±.0013 (4.0×)	.1372 (worse)	.0151±.0004	.0082±.0007 (7.6×)
cube (H2)	.0582±.0046	.0074±.0011 (7.9×)	.1346 (β_2 1→0)	—	—
torus (H1)	.0424±.0031	.0180±.0016 (2.3×)	.0457 (β_1 2→1)	.0155±.0015	.0133±.0004 (3.2×)
two spheres (H0)	.0065±.0008	.0007±.0002 (9.5×)	.0008 (ties C1)	—	—
double torus (H1)	.0264±.0000	.0264±.0000 (saturated)	.0264±.0000	—	—

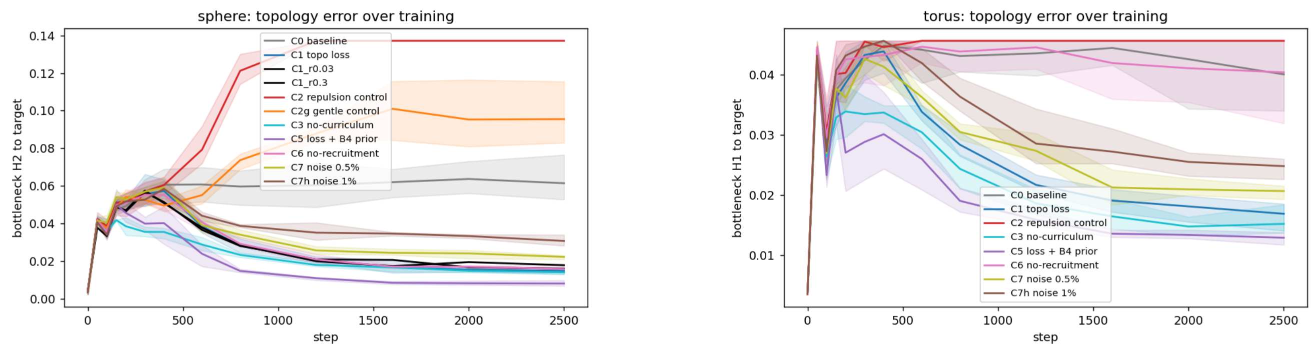

Figure 1. Topology error over training: bottleneck to target in the discriminating dimension (line = seed mean, band = seed min–max). Left, sphere (H2): every loss arm (C1, C3, C5, C6, both ρ variants, noise pair C7/C7h) converges to a fraction of baseline; the norm-matched repulsion control C2 is destructive, and its gentle variant C2g still ends worse than baseline. Right, torus (H1), the class no prior won: same ordering, C5 best; noise arms descend shallower but never approach baseline; the no-recruitment ablation C6 rides it.

or image complexity, so it amortizes where rendering dominates; no trainable parameters are added.

Recruitment is load-bearing for loops (C6). Removing the recruitment term (pure matching + diagonal, all else identical) collapses the torus win entirely: .0411±.0038, statistically baseline (0.6σ from C0, 12.4σ worse than C1) at baseline Chamfer — though the loops still exist (#sig H1 = 2 in every seed; Fig. 1, right: the C6 curve rides the baseline). The run logs locate the mechanism: one torus loop is significant at every refresh, the other chronically below threshold (all five seeds, all 200 refreshes) — matching and diagonal pressure act only on the measured loop, recruitment is the sole gradient path to the other; on this class it is the mechanism, not a cold-start device. On the sphere the void is always live and matched, recruitment never fires (0 of 200 refreshes, all seeds), so C6 is loss-identical to C1 by construction; its tail (.0166 vs .0156, 1.5σ) doubles as a measurement of pure run-to-run (CUDA) noise.

Sensor-noise stress (C7/C7h). With the plan's cloud perturbed at every refresh (σ = 0.5%/1% of the diagonal ≈ 0.5/1.0 median sample spacings, §4), the loss degrades gradually and never harms: torus .0210±.0018 / .0259±.0022 (2.0×/1.6× below baseline, vs C1's 2.3×), sphere .0235±.0009 / .0328±.0013 (2.7×/1.9×, vs 4.0×) — at equal-or-better Chamfer, with the correct significant-feature count in every run (Fig. 1). Internals: zero Gabriel-gate failures across all 2,400 noisy refreshes, and recruitment absorbed the noise (at σ=1% the torus's second loop is sub-threshold at every refresh; zero unreached targets). Calibration is the noisiest part (λ_{peak} spreads ≈3× across seeds), yet tails stay tight — consistent with the flat ρ response.

5.2. Generality: external genus-known meshes

Everything above is on one analytic family, so we repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ and ramp unchanged, density fixed by the floor rule (§4) — on four external genus-known meshes: spot (0, organic), bob (1, organic), fandisk (0, CAD), and the tom-yum pot (3, artist-authored, certified by ingest; §4). No per-shape tuning of any kind. All four pass the verdict rule at equal-or-better Chamfer (Table 2; trajectories in suppl. Fig. S2), and the control's failure modes replicate exactly: repulsion erases the void on all three H2 shapes (β_2 1→0, all seeds; on

Figure 2. The tom-yum pot: an artist-authored Blender model (raw export: 9 open shells, 232 non-manifold junction edges) ingested and certified to a single genus-3 shell, β=(1, 6, 1) exact (§4). Right: cutaway — the narrow flue mouth caps at small α, so the flue interior reads as an enclosed chamber: the on-axis H2 void the floor rule selects at M=8192 (all H1 loops sit below the floor).

the tom-yum pot at 1.36× worse Chamfer) and collapses a loop on the genus-1 shape (β_1 2→1, all seeds).

Four observations. The H1 claim survives the family change: bob halves loop error with zero phantom handles in every seed (26.1σ), replicating torus. Sharp-featured CAD benefits most: fandisk's 10.4× exceeds every analytic-family effect. Complex organic geometry exposes a count-level limit: spot's baseline manufactures a phantom void in two of three seeds (final #sig H2 = 2, 1, 2); the loss clears it in two seeds but retains one spurious bar in the third (1, 2, 1) — value repair is reliable, count repair on sub-budget organic detail (legs, horns at N=1200) is an improvement, not a guarantee; spot's C1 tail is pinned across seeds (sd 0 at .0237) — a geometric floor of the double-torus kind, far milder. And the pipeline reaches raw artist content: the tom-yum pot (Fig. 2) extends the wave in two directions at once — input class (a raw Blender export, certified rather than constructed, §4) and observable (an H2 void clearing the measurement floor only at M=8192) — both chosen by the pre-registered rule, not by hand: the dress rehearsal for scan data, whose ground truth must likewise be certified. The loss cuts the void's value error 3.9× at Chamfer parity (0.99× baseline); the 80.1σ Welch statistic reflects an unusually tight seed spread at n=3 (C1 sd .0002), so the honest summary is non-overlapping seed ranges, with the effect size inside the study's 2.1–10.4× span. The control destroys the void (β_2 1→0 in all three seeds) at 1.36× worse Chamfer; its bottleneck pins at the unmatched-target bound (the target bar's half-life, .0223), so the count — not the meaningless value gap (0.8σ) — is the verdict. Two caveats transfer: the baseline already reads the correct count (a value-accuracy win, like bob), and ground truth is certified, not constructed.

6. Mechanism and discussion

Sparse tugs vs. budget allocation. Each significant pair back-propagates through the 3–7 sample positions of its birth and death simplices: precise but sparse pulls, iterated by re-pairing — the toys show surgery "walking" around a defect a few points at a time; the resampling prior is the opposite — broad, indirect, budget-limited. The results sort cleanly along this axis: 1-D loops, which allocation can only over- or under-tessellate, respond to value pressure (H1 win); wide 2-D shells benefit from both (prior helps, loss helps more, composition best); 0-D structure needs neither (H0 null, three channels).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface: the void's death pins at that equilibrium (identical across seeds), then drags, while near-diagonal phantom bars accumulate (diagram trajectories: suppl. Fig. S1), and thin structures collapse (β_2 on the cube, β_1 on the torus). The loss has no length-scale constraint — it pulls the death coordinate itself and the geometry follows. So the control argument sharpens: the same gradient budget through the same channel without topological information is not neutral but harmful — C1's wins cannot be an artifact of extra regularization.

Measurability at the loss's density. A persistence loss can only enforce features that clear the significance floor at its sampling density M: the torus void and the double torus's two tube loops do not, at M=2048. We treat this as a principle, not a nuisance: the target bundle is built (and audited) at the same M, sub-floor features are excluded from the objective, and the double torus is evaluated against its two measurable loops — all the reconstruction itself can express at feasible budgets.

The floor is quantifiable: a bar is significant iff its lifetime exceeds $6\,r_{\text{med}}(M)$ (r_{med} = median nearest-neighbour spacing, the implementation's self-calibrating threshold), and r_{med} follows the uniform-sampling law $\propto M^{-1/2}$ (measured .0090→.0028 over M=2048→20000; ratio 3.16 vs $\sqrt{9.77}$=3.13). The double torus's tube loops live at lifetime .045–.046 on the clean surface: at M=2048 that is 0.83–0.85× the floor — invisible by a 15% margin, clearing it only from $M \approx 3{\times}10^3$ — while at the evaluation density (M=20000, floor .0171) they clear it 3.1×; the binding constraint there is the representation (the soup never expresses the tubes), so raising M alone cannot close the gap. The torus quantifies the interaction

Table 2. Generality wave: tail bottleneck to target (mean±sd, 3 seeds); × = reduction vs C0; σ = Welch. Same verdict rule as Table 1.

shape (dim)	C0	C1 loss	C2 control	verdict
spot (H2)	.0521±.0081	.0237±.0000 (2.2×)	.0652 (β_2 1→0)	PASS (6.1σ)
bob (H1)	.0426±.0012	.0208±.0008 (2.1×)	.0437 (β_1 2→1)	PASS (26.1σ)
fandisk (H2)	.0538±.0022	.0052±.0008 (10.4×)	.0571 (β_2 1→0)	PASS (35.7σ)
tom-yum pot (H2)	.0221±.0003	.0057±.0002 (3.9×)	.0223 (β_2 1→0)	PASS (80.1σ)

with noise: its second loop clears the clean floor by only 1.33×, a margin the live cloud's noise consumes — why it is chronically sub-threshold in training and recruitment is its only gradient path (C6). Recruitment is thus how the loss reaches below its own significance floor on the live side: landscape-guided in practice (unproductive tears self-extinguish at the local detour scale; suppl. §C), with the location-blindness caveat for target features beyond any achievable detour. Suppl. Table S3 tabulates the counts across M (at 4096 all four double-torus loops clear). The tom-yum pot ran the rule prospectively (§4), picking observable class and bundle density before any training run; the in-loop result (Table 2) validated it. Adaptive or multi-scale density is the natural extension; the trade is explicit, since persistence cost grows $O(M \log M)$.

Prescription. Correct topology in the loss; allocate with a wide prior; combine. Curriculum schedules are unnecessary once the loss scale is calibrated against the photometric gradient.

7. Limitations

Synthetic, closed, single-machine. All quantitative claims are on synthetic renders of closed surfaces with known topology, on one GPU: an analytic family plus four external genus-known meshes (§5.2; one a raw artist mesh, certified — the intended path for scans), short of a public benchmark. A real open-surface demonstration remains future work: scan boundaries contaminate H1 and no clean ground-truth diagram exists; of the concrete paths (boundary-aware persistence, filtering boundary-born bars, template-aligned initialization), bar-filtering is the pragmatic near-term step, a synthetic open-surface stress test the first experiment.

Density floor. The loss cannot see features below the significance floor at its sampling density (torus void; two of four double-torus loops at $M{=}2048$); the bundle audit makes this visible — but it is a real ceiling on what the objective can enforce.

Fixed target diagram. The target bundle presumes known ground-truth topology (a prior scan or template; not blind capture). Template-free designs are untested here: total-persistence minimization (risks collapsing legitimate structure), a target-genus penalty on the significant-H1 count (what we would try first), or statistical priors over a class's diagrams.

Control strength. The repulsion control is aggressive (it destroys topology at calibrated magnitude); a gentler variant at half radius is still 1.6× worse (sphere: .0972 vs .0623), so verdicts do not hinge on control tuning — but control design deserves more variety than one family at two strengths.

Nondeterminism. DiffSoup's CUDA training is not bit-reproducible: divergence enters at resampling, where borderline prune/split decisions flip under floating-point drift. Tail CVs run 7–10% for baseline and loss arms alike; a loss-identical re-run pair (the sphere C6 arm, C1 there by construction) landed 6% apart — pure run-to-run noise. All statements are seed-averaged with Welch statistics.

Recruitment guarantees. Recruitment's self-correction premise (unproductive edits saturate at the local detour scale) is validated empirically, not proven; pathological geometries (thin filaments, tightly constrained manifolds) could defeat it; a failure-case study and hardness analysis are future work.

8. Conclusion

We moved topology out of the resampler's allocation channel into the training objective. Four things carry it: a matched persistence loss whose pair-frozen backward is certified per pairing by a Gabriel-exactness gate; a recruitment term restoring the gradient optimal matching provably lacks for missing features (load-bearing for loops); channel-controlled evaluation (every gain beats a norm-matched non-topological control at Chamfer parity); and a quantified account of the floors (measurement, $6\,r_{\mathrm{med}}(M)$; representation, budget N). The verdicts replicate, without per-shape tuning, on external genus-known meshes (one a raw artist mesh certified by measurement) and degrade gracefully under noise. The prescription: correct topology in the loss, allocate with a wide prior, combine; next: open surfaces and real scans.

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